tf.concat

View source on GitHub (https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L1316-L1408)

Concatenates tensors along one dimension.

```
tf.concat(
    values, axis, name='concat'
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Better performance with the tf.data API (https://www.tensorflow.org/guide/data_performance) • Ragged tensors (https://www.tensorflow.org/guide/ragged_tensor) • TensorFlow basics (https://www.tensorflow.org/guide/basics) • Quickstart for the TensorFlow Core APIs (https://www.tensorflow.org/guide/core/quickstart_core) • Customizing what happens in \`fit()\` with TensorFlow (https://www.tensorflow.org/guide/keras/custom_train_step_in_tensorflow)	• Simple audio recognition: Recognizing keywords (https://www.tensorflow.org/tutorials/audio/simple_audio) • Learned data compression (https://www.tensorflow.org/tutorials/generative/data_compression) • Integrated gradients (https://www.tensorflow.org/tutorials/interpretability/integrated_gradients) • Load a pandas DataFrame (https://www.tensorflow.org/tutorials/load_data/pandas_dataframe) • Transfer learning for video classification with MoViNet (https://www.tensorflow.org/tutorials/video/transfer_learning_with_movinet)

See also `tf.tile` (https://www.tensorflow.org/api_docs/python/tf/tile), `tf.stack` (https://www.tensorflow.org/api_docs/python/tf/stack), `tf.repeat` (https://www.tensorflow.org/api_docs/python/tf/repeat).

Concatenates the list of tensors `values` along dimension `axis`. If `values[i].shape = [D0, D1, ... Daxis(i), ...Dn]`, the concatenated result has shape

```
[D0, D1, ... Raxis, ...Dn]
```

where

```
Raxis = sum(Daxis(i))
```

That is, the data from the input tensors is joined along the `axis` dimension.

The number of dimensions of the input tensors must match, and all dimensions except `axis` must be equal.

For example:

```
>>> t1 = [[1, 2, 3], [4, 5, 6]]
>>> t2 = [[7, 8, 9], [10, 11, 12]]
>>> tf.concat([t1, t2], 0)
<tf.Tensor: shape=(4, 3), dtype=int32, numpy=
array([[ 1,  2,  3],
       [ 4,  5,  6],
       [ 7,  8,  9],
       [10, 11, 12]], dtype=int32)>
```

```
>>> tf.concat([t1, t2], 1)
<tf.Tensor: shape=(2, 6), dtype=int32, numpy=
array([[ 1,  2,  3,  7,  8,  9],
       [ 4,  5,  6, 10, 11, 12]], dtype=int32)>
```

As in Python, the `axis` could also be negative numbers. Negative `axis` are interpreted as counting from the end of the rank, i.e., `axis + rank(values)`-th dimension.

For example:

```
>>> t1 = [[[1, 2], [2, 3]], [[4, 4], [5, 3]]]
>>> t2 = [[[7, 4], [8, 4]], [[2, 10], [15, 11]]]
>>> tf.concat([t1, t2], -1)
<tf.Tensor: shape=(2, 2, 4), dtype=int32, numpy=
  array([[[ 1,  2,  7,  4],
          [ 2,  3,  8,  4]],
         [[ 4,  4,  2, 10],
          [ 5,  3, 15, 11]]], dtype=int32)>
```

★ **Note:** If you are concatenating along a new axis consider using stack. E.g.

```
tf.concat([tf.expand_dims(t, axis) for t in tensors], axis)
```

can be rewritten as

```
tf.stack(tensors, axis=axis)
```

Args	
`values`	A list of `Tensor` objects or a single `Tensor`.
`axis`	0-D `int32` `Tensor`. Dimension along which to concatenate. Must be in the range `[-rank(values), rank(values))`. As in Python, indexing for axis is 0-based. Positive axis in the rage of `[0, rank(values))` refers to `axis`-th dimension. And negative axis refers to `axis + rank(values)`-th dimension.
`name`	A name for the operation (optional).

Returns
A `Tensor` resulting from concatenation of the input tensors.